SIMATS ENGINEERING

LAB REPORT

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Course Name:	Knowledge Engineering and Data Science
Course Code	DSA1605
Lab Details:	Exp 1 To 3

Exp 1 :

1. Pie Chart and Bar Chart of Student Grades

Question: Write a Python program to display student grades using a pie chart and bar chart.

Aim: To represent the percentage distribution and number of students across different grades using charts.

Procedure:

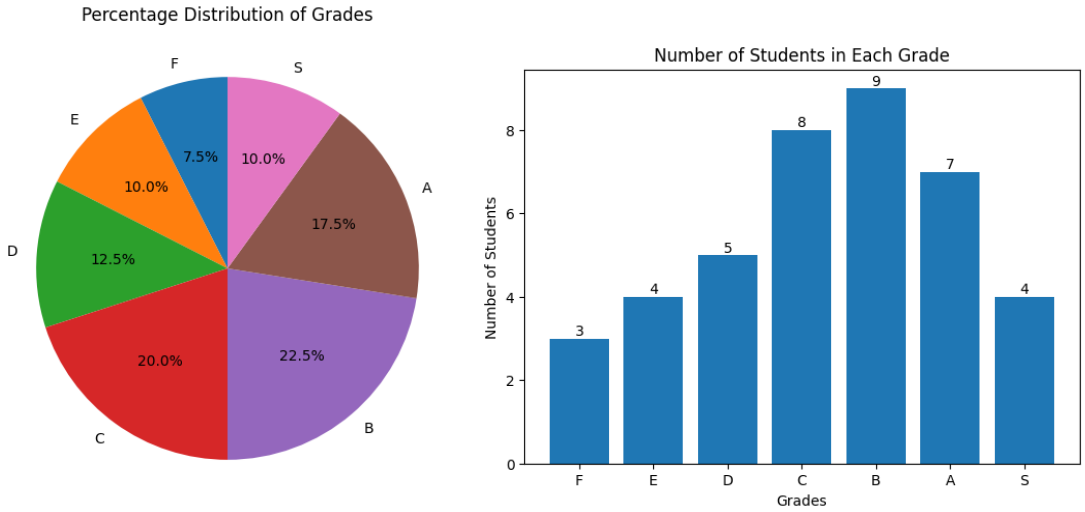
1. Store grades and student counts in lists.
2. Draw a pie chart with percentage values.
3. Draw a bar chart with student counts.
4. Find maximum and minimum grades.
5. Calculate the combined percentage of grades A and S.

CODE:

```
import matplotlib.pyplot as plt
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]
plt.pie(students, labels=grades, autopct='%1.1f%%')
plt.title("Grade Distribution")
plt.show()
plt.bar(grades, students)
plt.xlabel("Grades")
plt.ylabel("Number of Students")
plt.title("Student Grades")
for i in range(len(grades)):
    plt.text(i, students[i], students[i], ha='center')
plt.show()
print("Maximum Grade:", grades[students.index(max(students))])
print("Minimum Grade:", grades[students.index(min(students))])
```

print("A and S Percentage:", (7 + 4) / 40 * 100, "%")

Output :



Maximum students: B - 9

Minimum students: F - 3

Percentage of A and S grades: 27.500000000000004 %

Maximum students: ['B'] - 9

Minimum students: ['F'] - 3

Students securing A and S: 11

Combined percentage of A and S: 27.5 %

Result:

The pie chart and bar chart were generated successfully. Grade B has the maximum students (9), grade F has the minimum (3), and grades A and S together account for 27.5%.

Exp : 2

2. Semantic Network Using NetworkX

Question: Write a Python program to represent a semantic network using NetworkX.

Aim: To represent concepts and their relationships using a directed graph.

Procedure:

1. Import the NetworkX library.
2. Create a directed graph.
3. Add concepts and labeled relationships.
4. Display all relationships.

Code:

```
import networkx as nx
G = nx.DiGraph()
G.add_edge("Dog", "Mammal", relation="is-a")
G.add_edge("Mammal", "Animal", relation="is-a")
G.add_edge("Dog", "Tommy", relation="has-instance")
print("Semantic Network Relationships:")

for s, t, d in G.edges(data=True):
    print(s, "--", d["relation"], "-->", t)
```

Output:

Semantic Network Relationships:

Dog -- is-a --> Mammal

Dog -- has-instance --> Tommy

Mammal -- is-a --> Animal

Result

The semantic network was created successfully, representing concepts and their relationships.

Exp 3:

3. Frame-Based Knowledge Representation

Question: Write a Python program to represent knowledge using frames.

Aim: To represent an object using a frame containing slots and their values.

Procedure:

1. Create a dictionary to represent a frame.
2. Store attributes as keys and values as slot values.
3. Display the frame and its attributes.

Code:

```
vehicle = {
    "class": "Vehicle",
    "brand": "Toyota",
    "fuel_type": "Petrol",
    "wheels": 4,
    "color": "White"
}
print("Vehicle Frame")
for slot, value in vehicle.items():
    print(slot, ":", value)
```

Outcome:

Vehicle Frame

class : Vehicle

brand : Toyota

fuel_type : Petrol

wheels : 4

color : White

Result:

Thus, the vehicle was successfully represented using a frame containing slots and their corresponding values.